

Pi Camera

DETECT

IMAGES

VIDEO

What does it do?

A camera turns light into digital images. Connect a Raspberry Pi Camera Module to a Raspberry Pi computer with a flat ribbon cable.

Use Python to take a photograph or record a video. Your program can also examine images and decide what to do next.

Uses



Video doorbell

DETECT

Camera sees movement.

DECIDE

Is a person there?

DO

Save a video clip.



Monster photo booth

DETECT

Button starts countdown.

DECIDE

Has time run out?

DO

Take a silly photo.



Lens

Focuses light

Sensor

Records the image

Ribbon

Connects to the Pi

Code

● Take photo

```
from picamera2 import Picamera2

camera = Picamera2()
camera.start_and_capture_file(
    "photo.jpg", delay=2,
    show_preview=False)
camera.close()
```

● Take video

```
from picamera2 import Picamera2

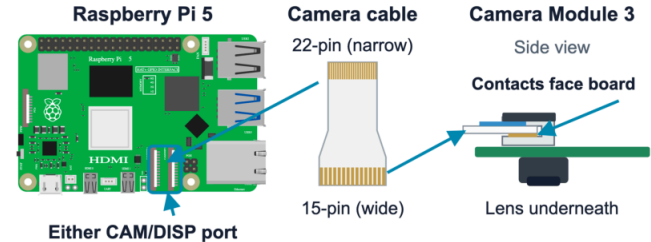
camera = Picamera2()
camera.start_and_record_video(
    "video.mp4", duration=5)
camera.close()
```

Run each example separately on **Raspberry Pi OS** with **Picamera2** installed. Files save in the current folder. The video is five seconds long, without sound.

Wiring

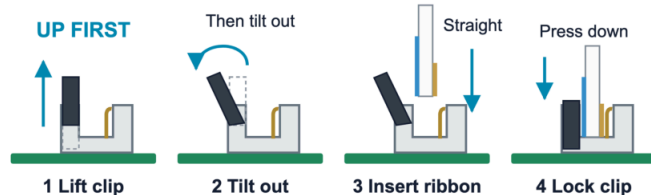
RASPBERRY PI 5 + CAMERA MODULE 3

POWER OFF • Unplug USB-C before fitting



Lift UP first → tilt → insert → lock

Pi 5 connector shown from the side



Blue backing faces the flap; metal contacts face away.

At the camera, contacts face towards the circuit board.

Seat both edges evenly. Do not force or crease the ribbon.

Use a **Standard-Mini camera cable** (22-pin to 15-pin). The wide cable supplied with the camera fits a Pi 4, not a Pi 5.